

DEFECTS IN ANYON SYSTEMS

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Abstract

Systems with topological order (TO) in 2+1D have exotic excitations over the ground state, which we call anyons. With Kitaev's proposal that we can use non-abelian anyons for topological quantum computation (TQC), there has been a boom in this field. With these anyons (topological quasiparticles), we can perform different operations like- braiding and fusion. The initial part of the thesis builds up the basic ideas of anyons, their field theory, Wilson loops, simple anyon models, and doing TQC with anyons. Later, we look into twist defects (extrinsic defects) in anyon systems where quasiparticle type changes under some symmetry after going around the twist defect. These extrinsic defects have a non-trivial quantum dimension which is the signature of non-abelian topological quasiparticles though they are not exactly like anyons. These extrinsic defects follow projective non-abelian statistics. Some specific examples of such twist defects have been studied in the toric code model and the bilayer fractional quantum Hall (FQH) system. The twist defects in the bilayer FQH systems can be mapped smoothly to a higher genus surface, hence called Genons. These twist defects are of interest because they have the potential to be used for universal TQC as they behave in some sense similar to non-abelian anyons.

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1. Introduction

Anyons are particles found in 2d systems which show $\psi(\mathbf{r}_1, \mathbf{r}_2) = e^{i\theta} \psi(\mathbf{r}_2, \mathbf{r}_1)$ when particles are exchanged. The possibility of such statistics arising was shown by Leinaas & Myrheim [1], and their properties were studied by Wilczek [2, 3]. In the below sections, these ideas are explained briefly, and as a concrete model to understand the properties of anyons, the Toric code model has been introduced, followed by how these systems can be used for robust quantum computation.

1.1 Anyons: Fractional statistics in 2D

In quantum mechanics, we care about paths and path integration. So when asked the question of whether we should exchange particles just by exchanging their positions in the wavefunction or should we adiabatically transport particles for calling it an exchange. In three dimensions, both the above statements are equivalent, whereas, in two spatial dimensions, we take the latter definition.

The adiabatic exchange of identical particles in 3d can be thought of as going from a point $\mathbf{r}$ in relative configuration space to either $\mathbf{r}$ or $-\mathbf{r}$. So all such paths can be continuously deformed and put into two such inequivalent paths, i.e., $\Pi_1((\mathbb{R}_3 - origin)/\mathbb{Z}_2) = \mathbb{Z}_2$, and for 2d, the number of such inequivalent paths is $\Pi_1((\mathbb{R}_2 - origin)/\mathbb{Z}_2) = \mathbb{Z}$.

The act of exchanging particles in 2d can be described in terms of braid group $\mathbb{B}_N$ [†] and parallelly in 3d by the permutation group $\mathbb{P}_N$. The braid group $\mathbb{B}_N$ generators are σ_j , where $j = 1, 2, \dots, N - 1$; and σ_j denotes the exchange of j and $j+1$ particle. The generators satisfy the below defining relations:

$$\begin{aligned} \sigma_i \sigma_j &= \sigma_j \sigma_i, \quad |i - j| \geq 2 \\ \sigma_j \sigma_{j+1} \sigma_j &= \sigma_{j+1} \sigma_j \sigma_{j+1} \quad (\text{Yang - Baxter relation}) \end{aligned} \tag{1.1}$$

[†]The general references for the introduction and, in general, for the whole thesis are [4, 5, 6, 7, 8]

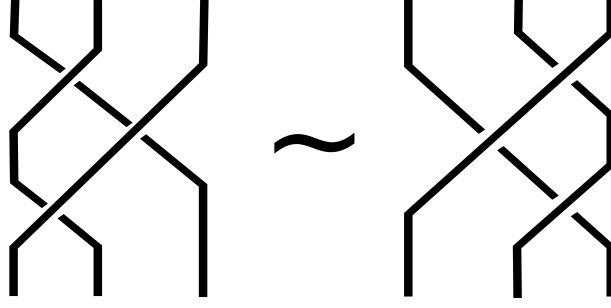


Figure 1.1: If we keep both the ends of strings fixed, then one can smoothly change one to the other, i.e., $\sigma_j \sigma_{j+1} \sigma_j = \sigma_{j+1} \sigma_j \sigma_{j+1}$

It can be seen easily that the one-dimensional representation of $\mathbb{B}_N$ is $\sigma_j = e^{i\theta} \forall j$, giving us the abelian anyons.

1.2 Abelian anyons in Toric code

The Hamiltonian of the toric code model [9] is given as

$$H = -J_e \sum_{\nu} A_{\nu} - J_m \sum_p B_p \quad (1.2)$$

where

$$A_{\nu} = \prod_{j \in \nu} \sigma_x^j \quad \text{and} \quad B_p = \prod_{j \in p} \sigma_z^j. \quad (1.3)$$

Some important observations of the plaquette and vertex operators are:

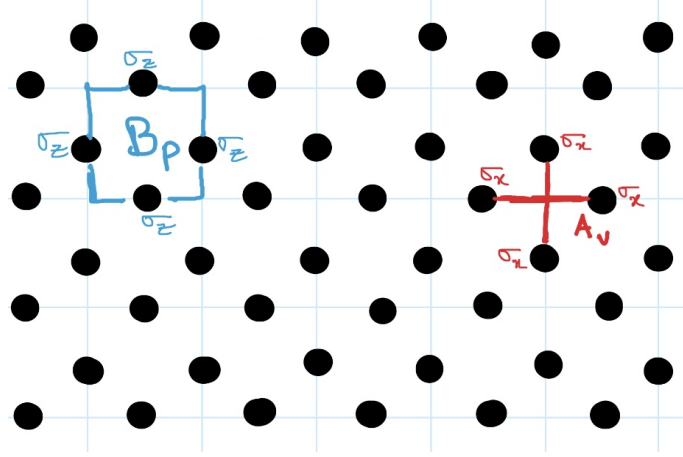
$$[A_{\nu}, A_{\nu'}] = [B_p, B_{p'}] = [A_{\nu}, B_p] = 0 \quad \forall \quad \nu, \nu', p, p' \quad (1.4)$$

We first find the GS of either the plaquette term or the vertex term. Then observe that an equal superposition of all such configurations turns out to be the GS of the Hamiltonian.

$$|\Psi_0\rangle = N \prod_v (1 + A_v) |\zeta\rangle, \quad (1.5)$$

where $|\zeta\rangle$ is some reference state like all spins pointing up in z-direction. The energy of the GS is $E = -2N^2 J_e - 2N^2 J_m$.

To get the excited states, we do $\sigma_x^i |\Psi_0\rangle$ or $\sigma_z^i |\Psi_0\rangle$, which raises the GS energy by $4J_m$ or $4J_e$, respectively. The excitation causes a pair of B_p or A_v eigenvalues to flip from $=1$ to -1 . These pairs can freely move around without a change in energy. What happens if we make them go around a loop? These loops don't change any of B_p or A_v eigenvalues



and can be seen as commuting with the Hamiltonian. One observes that the GS has the degeneracy of 4 if this lies on a torus as on a torus we can have non-contractible loops. The other excitation we can create is $i\sigma_y^i|\Psi_0\rangle = \sigma_z^i\sigma_x^i|\Psi_0\rangle$. This creates a pair of e and m excitations; we call this excitation $\epsilon = e \times m$. These quasiparticles have the below fusion and braiding rules:

$$e \times e = I, \quad m \times m = I, \quad \epsilon \times \epsilon = I, \quad e \times m = \epsilon, \quad e \times \epsilon = m, \quad m \times \epsilon = e \quad (1.6)$$

$$R_{ee} = R_{mm} = 1, \quad R_{\epsilon\epsilon} = -1, \quad R_{em} = i. \quad (1.7)$$

1.3 Universal Topological Quantum Computation with Anyons

1.3.1 Universal Quantum Computation

Suppose we take the Hilbert space of N qubits, whose Hilbert space dimension is $D = 2^N$. Then the space of all possible unitary operations on these qubits forms a group of $D \times D$ unitary matrices written as $U(D)$.

Let the quantum computer has some finite number- p of gates (most elementary operations), which it can implement at each timestep. Then we denote each of the gates as $U_n \in U(D)$, where $n \in 1, \dots, p$. A sequence of such gates constructs a particular unitary operation which is just the matrix multiplication of those gates written as

$$U = U_{i_t} \dots U_{i_2} U_{i_1} \quad (1.8)$$

where the number of gates t can be thought of as the running time for the computation. The idea of universal quantum computation (UQC) is that we can construct (up to certain accuracy) any desired unitary operation from the repeated use of some combinations of

those p set of gates. To clarify the notion of what we meant by up to some accuracy, we define the distance measure between two unitary matrices as

$$\mathbf{dist}(U; V) = \sqrt{1 - \frac{|\mathrm{Tr}[U^\dagger V]|}{D}}. \quad (1.9)$$

It is easy to see that $\mathbf{dist}(U; U) = 0$, and $\mathbf{dist}$ is invariant if we multiply the matrix by an external phase factor. So, the **formal definition for UQC** is that for any matrix $U_{\text{target}} \in U(D)$ and any $\epsilon > 0$, we can find a sequence of gates such that its distance from the target matrix is less than any desired error tolerance ϵ , i.e.,

$$\mathbf{dist}(U_{\text{target}}; U_{i_t} \dots U_{i_2} U_{i_1}) < \epsilon. \quad (1.10)$$

A theorem by Kitaev and Solovay gives an algorithm by which one is guaranteed to find such a sequence that is not very long ($t \sim \mathcal{O}(\log(1/\epsilon))$). An important non-trivial calculation then is to determine which sets of gates are sufficient to have a quantum computer that is universal. We have the Brylinski theorem for our rescue, which says that if we can perform arbitrary single-qubit rotation and also perform any two-qubit entanglement operation between any of these two bits (or even just the neighboring bits), then we have our universal quantum computer.

1.3.2 Topological Quantum computing using anyons

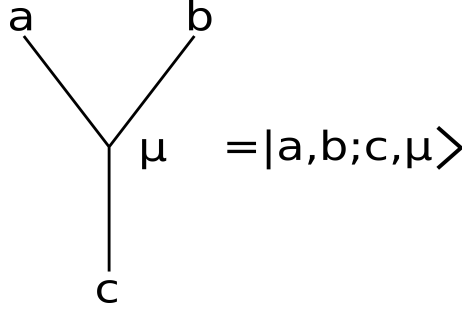
With anyons, we can have only two operations: fusion and braiding. We define particle types with a finite set of labels- a, b, c, etc. All anyon systems have an identity particle or vacuum denoted by I or e .

Worldlines of particles can merge, which we call fusion, and the reverse process is called splitting. If particles a and b merge to give a particle c, we denote it as $a \times b = b \times a = c$. See the Fig... for a diagrammatical representation of fusion.

In our abelian anyon model of charges and fluxes, if the statistical angle $\theta = \pi p/m$, then we have m different types of anyons denoted by $a = (aq, a\Phi)$, where $a = 0, 1, \dots, m-1$ and $q\Phi = \pi p/m$. Fusion rules are simply addition modulo m , i.e., $a \times b = (a+b) \bmod m$. The fusion of non-abelian anyons can have more than one possible result, like $a \times b = c+d$. Also, we can get the same particle through different fusion channels. So, in general, we write fusion of two particles give

$$a \times b = \sum_c N_{ab}^c, \quad (1.11)$$

where in general for all our models of interest will have $N_{ab}^c = 1$ or 0. If two particles

Figure 1.2: Fusion diagram of μ channel where $\mu \in \{1, 2, \dots, N_{ab}^c\}$

fuse and one of the fusion channels give the result of fusion as an I , then we have the particle antiparticle pair. Note that antiparticles are unique and a particle can be it's own antiparticle.

We write the fusion information N_{ab}^c in the N matrices. The N_a matrix for an anyon a is just the matrix $[N_a]_b^c$. The N matrices for different anyons commute because of the associativity of fusion, and the quantum dimension for an anyon type a is the largest eigenvalue of the N_a matrix (See Appendix B.2).

1.3.3 R-matrices & F-matrices

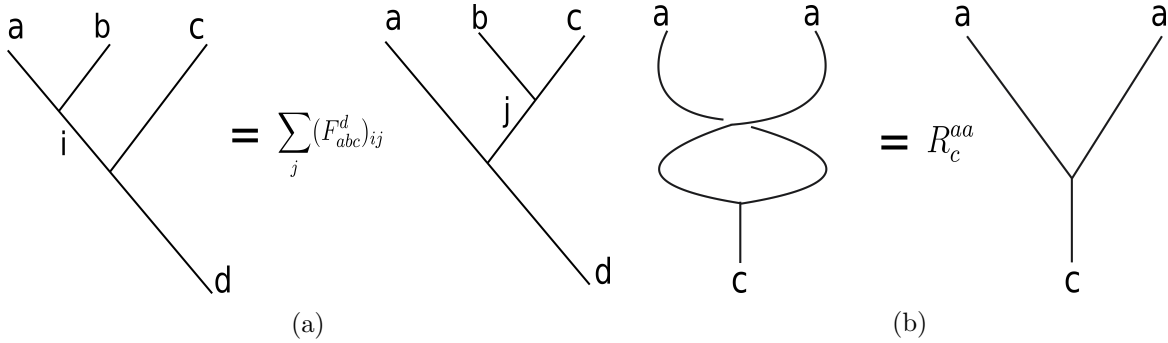


Figure 1.3: (a) F-matrix tree diagram (b) R-matrix tree diagram

As shown in fig(1.3.a) we can describe the change of basis in that diagram from one vector to another by a set of unitary matrices called F matrices. The F matrix in the figure is labelled as F_d^{abc} . The F matrix has to follow a constraint known as the Pentagon equation which can be justified from the diagram fig(1.4). Coherence theorem by MacLane guarantees that all other diagrams that can be constructed enforce no more constraints than the pentagon equations.

When we exchange two particles in a fusion tree (as shown in fig(1.3.b)) we get a phase term denoted by R_c^{aa} . If we assign basis vectors as in fig(1.5), we can just multiply

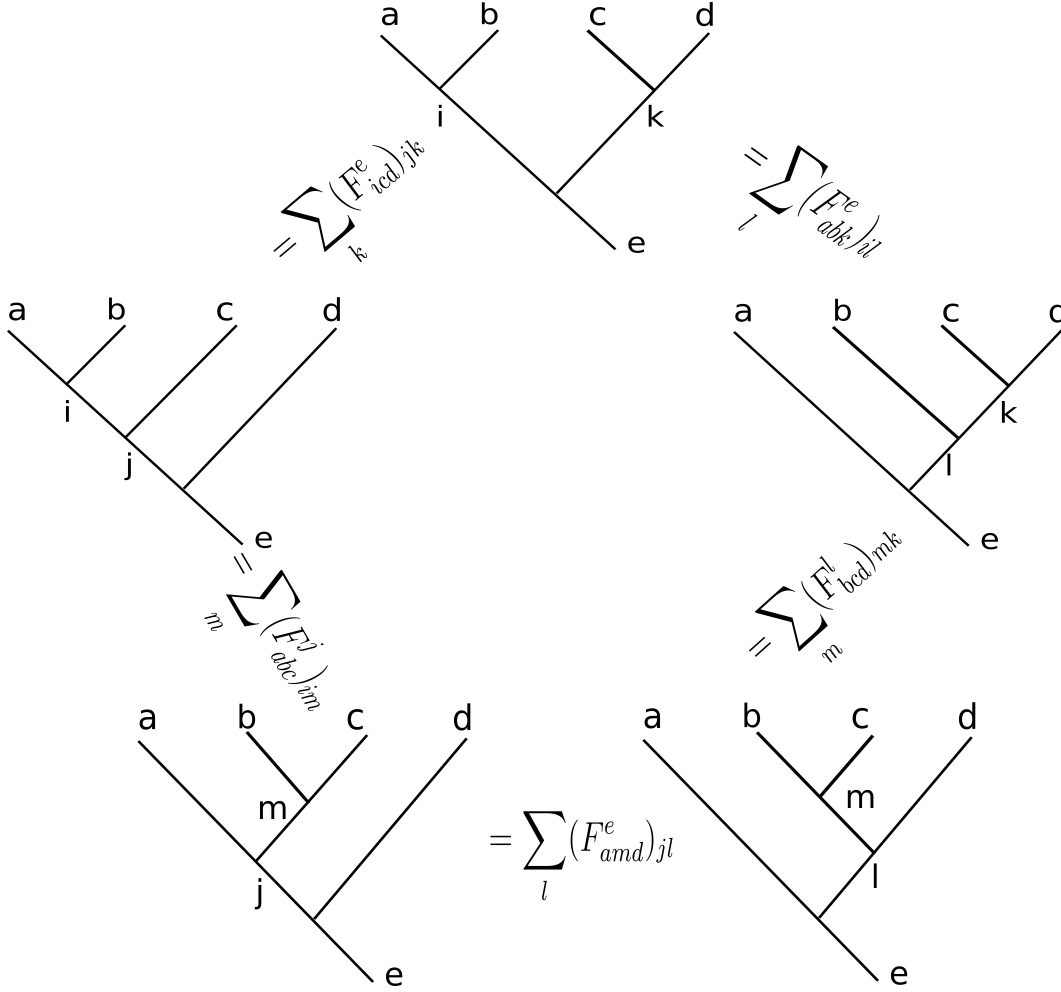


Figure 1.4: The Pentagon diagram which enforces constraints on the F-matrix.

those vectors by a phase term if we exchange the first two anyons (σ_1 operation). The difficulty arises if we operate σ_2 on these diagrams. The idea then is to use F-matrix to go to the basis where second and third particles fuse, then apply the R-matrix and then apply the inverse F-matrix to come back to the same basis.

1.3.4 Examples: Fibonacci and Ising anyons

Fibonacci anyons

The anyons are denoted by the symbols $1, \tau$, and the fusion rules for these are

$$1 \times \tau = \tau \quad \tau \times \tau = 1 + \tau. \quad (1.12)$$

As we can see above that the $\tau \times \tau$ fusion channel gives more than one topological quasiparticle. Hence, these are non-abelian anyons. Fibonacci anyons are the simplest non-abelian anyon system. Note, one can perform UTQC using Fibonacci anyons. These

are predicted to be found in $\nu = 12/5$ FQH system and are described by $SU(2)_3$ Chern-Simons theory. As one can see from the fusion tree, the quantum dimension of the τ particles scales as ϕ^N for a system of N particles. Hence, we say that the quantum dimension for Fibonacci anyons is $\phi = \frac{1+\sqrt{5}}{2}$. The N_τ matrix is given as

$$N_\tau = \begin{pmatrix} I & \tau \\ I & 0 & 1 \\ \tau & 1 & 1 \end{pmatrix} \quad (1.13)$$

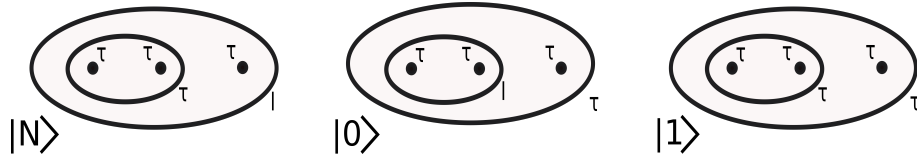


Figure 1.5: The three different fusion channels for three Fibonacci anyons

Ising anyons

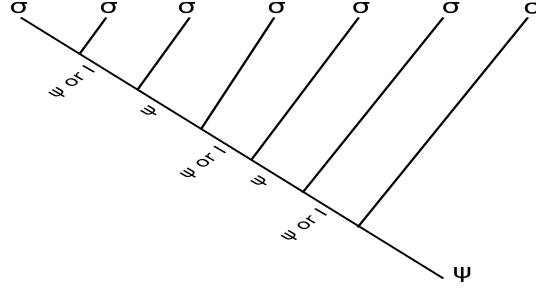
The particles type for Ising anyons are: $\{I, \sigma, \psi\}$. The fusion rules are given by

$$\psi \times \psi = I \quad \psi \times \sigma = \sigma \quad \sigma \times \sigma = I + \psi \quad (1.14)$$

The ψ anyons never fuse to give more than one particle type and are hence abelian. Whereas the σ anyons are non-abelian, and their N_σ matrix is given by

$$N_\sigma = \begin{pmatrix} I & \sigma & \psi \\ I & 0 & 1 & 0 \\ \sigma & 1 & 0 & 1 \\ \psi & 0 & 1 & 0 \end{pmatrix} \quad (1.15)$$

Hence, the quantum dimension for σ anyons, $d_\sigma = \sqrt{2}$. These are predicted in $\nu = 5/2$ FQH systems, topological superconductors and are closely related to Majorana systems and $SU(2)_2$ Chern-Simons theory.

Figure 1.6: Ising model fusion tree for 7 σ anyons

1.4 Low energy EFT of such models

The low energy effective field theory of systems with topological order follows Topological Quantum Field Theory (TQFT [10]). There are different ways to define TQFTs; one is that the field theory must be independent of small deformations of the space-time metric, i.e.,

$$\frac{\delta}{\delta g_{\mu\nu}} \langle \text{any correlator} \rangle = 0. \quad (1.16)$$

The other definition of a TQFT is that it's a functor¹ from cobordism category **nCob** to the category of Hilbert spaces **Hilb**. To understand these words more clearly, let's follow this TQFT definition from Atiyah's axioms [11] which was inspired by Segal's axioms for conformal field theory.

The space-time manifold $\mathcal{M}$ considered in the below discussion is $D+1$ dimensional, and Σ is a D dimensional oriented space slice at a fixed point. Below is a rough interpretation of Atiyah's axioms for physicists. To begin with, I have considered space-time manifolds with no particles in them. One can extend these for particles by thinking of particle worldlines as holes in the manifold $\mathcal{M}$ which acts as a manifold boundary.

AXIOM 1: A D dimensional space Σ is associated with a Hilbert space $V(\Sigma)$, which only depends on the topology of Σ .

AXIOM 2: The disjoint union of two D -dimensional spaces Σ_1 and Σ_2 will be associated with a Hilbert space which is a tensor product of the two Hilbert spaces, i.e.,

$$V(\Sigma_1 \cup \Sigma_2) = V(\Sigma_1) \otimes V(\Sigma_2).$$

As a corollary to axiom 2, we can see that since $\Sigma \cup \emptyset = \Sigma$, so $V(\emptyset) = \mathbb{C}$.

AXIOM 3: For a $D+1$ dimensional manifold $\mathcal{M}$, if Σ is the D dimensional boundary, $\Sigma = \partial\mathcal{M}$. Then we attach a particular element of the vector space $V(\Sigma)$ with this manifold

$$Z(\mathcal{M}) \in V(\partial\mathcal{M}).$$

AXIOM 4: Reversing orientation gives the dual space of the associated vector spaces, i.e., $V(\Sigma^*) = V^*(\Sigma)$.

GLUING: We can glue two manifolds $\mathcal{M}$ and $\mathcal{M}'$ if they have a common boundary $\Sigma = \partial\mathcal{M} = (\partial\mathcal{M}')^*$, and that makes the new vector associated with the glued manifold

¹See Appendix A.1

as $Z(\mathcal{M} \cup_{\Sigma} \mathcal{M}') = \langle Z(\mathcal{M}') | Z(\mathcal{M}) \rangle$.

COBORDISM: If we can find a manifold $\mathcal{M}$ such that the boundary of the manifold is the disjoint union of two manifolds Σ_1 and Σ_2 , i.e., $\partial\mathcal{M} = \Sigma_1 \cup \Sigma_2$. Then $\mathcal{M}$ is called a cobordism between Σ_1 and Σ_2 .

If we have a manifold $\mathcal{M}$, which is a cobordism such that $\partial\mathcal{M} = \Sigma_1^* \cup \Sigma_2$, then $Z(\mathcal{M}) \in V(\Sigma_1^*) \otimes V(\Sigma_2)$. Hence, we can write $Z(\mathcal{M}) = \sum_{\alpha\beta} U^{\alpha\beta} |\psi_{\Sigma_2, \alpha}\rangle \otimes \langle \psi_{\Sigma_1, \beta}|$.

Effective Field Theory of Fractional Quantum Hall systems

After the discovery of IQHE in 1980, the experiments done by Tsui et al. [12] in 1982 showed peaks at fractional filling. For the FQHE to show up, one needs to have a high magnetic field and a very clean system. Everyone expected to get Wigner crystals at high magnetic fields (low filling fractions), but it turned out that electron-electron interaction causes the system to order up in a beautiful way and form an in-between liquid phase before forming Wigner crystals. The IQHE ground state is highly degenerate; because of the e-e interaction, the degeneracy is lifted, and at those fractional fillings, a gap arises, which explains the observed effects like FQHE.

The Effective field theory for general abelian FQH systems is of the form below [13] :

$$\mathcal{L} = \frac{1}{4\pi} K_{IJ} a_{I\mu} \partial_{\nu} a_{J\lambda} \epsilon^{\mu\nu\lambda} - e A_{\mu} T_I \partial_{\nu} a_{I\lambda} \epsilon^{\mu\nu\lambda} \quad (1.17)$$

For multilayer FQH systems, the K-matrix has all integer entries, and the diagonal entries are odd integers. Also, the charge vector is given by $t^{\mathbf{T}} = (1, 1, \dots, 1)$.

For the n^{th} level hierarchical states the

$$\text{K-matrix : } \begin{pmatrix} p_1 & -1 & 0 & 0 & \dots \\ -1 & p_2 & -1 & 0 & \dots \\ 0 & -1 & p_3 & -1 & \dots \\ \vdots & & & \ddots & \vdots \\ 0 & \dots & & -1 & p_n \end{pmatrix} \text{ and } t = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}.$$

Wen [14] showed that we could characterize different FQH states by $(K, \mathbf{t})$ such that two states are equivalent if one can find the below transformation between them:

$$\mathbf{t}_2 = W \mathbf{t}_1, \quad K_2 = W K_1 W^T \quad (1.18)$$

where $W \in SL(n, \mathbb{Z})$.²

²Note that here we ignored a topological quantum number- spin vector. This doesn't cause any problem if we have disordered FQH systems where the spin vector is not well defined

2. Chern-Simons field theory

The abelian Chern-Simons (CS) theory action is given by

$$S_{CS}[A] = \frac{k}{4\pi} \int_{\mathcal{M}} d^3x \epsilon^{\mu\nu\lambda} A_\mu \partial_\nu A_\lambda \quad (2.1)$$

where A 's are the gauge fields and k is called the level of the CS term. The CS term is rotational invariant but breaks both parity and time symmetries.

2.1 Classical Chern-Simons theory

The pure Chern-Simons theory[†] eq.(2.1) in itself is boring. We can couple this field with matter degrees of freedom to make it more interesting.

$$S_{CS}[A] = \frac{k}{4\pi} \int_{\mathcal{M}} d^3x [\epsilon^{\mu\nu\lambda} A_\mu \partial_\nu A_\lambda - A_\mu J^\mu] \quad (2.2)$$

We find the resulting equations of motion by taking, $\delta S_{CS} = 0$.

$$\implies \frac{k}{2\pi} \epsilon^{\mu\nu\lambda} \partial_\nu A_\lambda = J^\mu \quad (2.3)$$

Writing the matter current as $J^\mu \equiv (\rho, \mathbf{J})$ and the four vector notation we use is as follows: $A^\mu \equiv (\phi, \mathbf{A})$; $\partial_\mu \equiv (\frac{d}{dt}, \nabla)$ and the metric tensor is $\text{diag}(-1, +1, +1)$. Since,

$$B = \nabla \times \mathbf{A} \text{ and } E_x = -\frac{d\phi}{dt} - \frac{dA_x}{dx}, \text{ similarly for } E_y.$$

So, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ implies $F_{\mu\nu} = \begin{pmatrix} 0 & E_1 & E_2 \\ -E_1 & 0 & B \\ -E_2 & -B & 0 \end{pmatrix}$. Hence the equations of motion

become

$$\rho = \frac{k}{2\pi} B \quad \text{and} \quad J^i = \frac{k}{2\pi} \epsilon_{ij} E_j. \quad (2.4)$$

[†]The main references for this chapter are [5, 15, 16]

From this one can say that the charge density is locally proportional to the magnetic field, i.e., the Chern-Simons term binds a magnetic flux to a charge. Now setting the particles as point particles then the charge density becomes

$$\rho(\vec{x}, t) = e \sum_{a=1}^N \delta^{(2)}(\vec{x} - \vec{x}_a(t)) \quad (2.5)$$

and current density is

$$\vec{j}(\vec{x}, t) = e \sum_{a=1}^N \dot{\vec{x}}_a(t) \delta^{(2)}(\vec{x} - \vec{x}_a(t)) \quad (2.6)$$

Moving the charges adiabatically (extremely slowly) we can set $\mathbf{J} = 0$, and set the Coulomb gauge, i.e., $\partial_i A^i = 0$ and put $A_0 = 0$. The Coulomb gauge and eq(2.3) give

$$\nabla^2 A_x = -\frac{2\pi}{k} \partial_2 \rho \quad \text{and} \quad \nabla^2 A_y = \frac{2\pi}{k} \partial_1 \rho. \quad (2.7)$$

In two dimension we have the below Green's function for the Laplacian

$$\nabla^2 \left(\frac{1}{2\pi} \log |\vec{x} - \vec{y}| \right) = \delta^{(2)}(\vec{x} - \vec{y}). \quad (2.8)$$

This implies

$$A_x = -\frac{2\pi}{k} \int \left(\frac{1}{2\pi} \log |\vec{x} - \vec{y}| \right) \partial_2 \rho(\vec{y}, t) d^2 y. \quad (2.9)$$

Using integration by parts (IBP) and putting the first term as zero, because charge density vanishes at infinity.

$$\Rightarrow A^i(\vec{x}, t) = -\frac{e}{k} \sum_{a=1}^N \epsilon^{ij} \frac{(x^j - x_a^j(t))}{|\vec{x} - \vec{x}_a(t)|^2} \quad (2.10)$$

Using the identity $\partial_i \arg(\vec{x}) = -\epsilon_{ij} x^j / |\vec{x}|^2$, where $\arg(\vec{x}) = \arctan(\frac{x^2}{x^1})$ implies

$$A_i(\vec{x}, t) = \frac{e}{k} \sum_{a=1}^N \partial_i \arg(\vec{x} - \vec{x}_a(t)). \quad (2.11)$$

This gauge vector potential is like a total derivative, hence the fields can be thought of as zero now but doing such a gauge transform makes our wavefunction go as

$$\psi(\vec{x}) \rightarrow \tilde{\psi}(\vec{x}) = \exp \left(-i \frac{e^2}{k} \sum_{a=1}^N \arg(\vec{x} - \vec{x}_a) \right) \psi(\vec{x}) \quad (2.12)$$

It looks like the gauge field has been gauged away, and all we have is a free system but it turns out there is a complex statistical interaction which is hidden in the non-trivial boundary conditions for the non-single-valued field $\tilde{\psi}$. For Lorentz gauge the solution is

(see Appendix B.1)

$$A_\alpha = \frac{1}{2k} \int d^3y \frac{\epsilon_{\alpha\beta\mu} \partial^\beta J^\mu}{|x - y|} \quad (2.13)$$

2.2 Quantising abelian Chern-Simons theory

We see what happens to the action under the gauge transform $A_\mu \rightarrow A_\mu + \partial_\mu \omega$,

$$S_{CS}[A] \rightarrow S_{CS}[A] + \frac{k}{4\pi} \int_{\mathcal{M}} d^3x \partial_\mu (\omega \epsilon^{\mu\nu\lambda} \partial_\nu A_\lambda). \quad (2.14)$$

The action is gauge invariant if ω vanishes at the boundary or if all the fields ($F_{\mu\nu}$) vanish at the boundary. We push this invariance to the extreme by compactifying the space-time manifold $\mathcal{M} = S^2 \times S^1$ and set ω not vanishing at boundaries and put a magnetic monopole inside S^2 . The magnetic monopole produces non-zero flux on S^2 , which we set to it's minimum value possible by the Dirac quantization condition

$$\int_{S^2} d^2x F_{12} = 2\pi. \quad (2.15)$$

We choose the gauge transformation $\omega(\tau, \mathbf{x}) = \frac{2\pi\tau}{\beta}$, where β is the radius of the compactified circle, S^1 and now $A_0 \rightarrow A_0 + \frac{2\pi}{\beta}$. So,

$$S_{CS} = \frac{k}{4\pi} \int d^3x (A_0 F_{12} + A_1 F_{20} + A_2 F_{01})$$

. We observe that only A_0 changes by our above defined gauge transform, but A_0 is also in F_{20} and F_{01} which if moved to the form $A_0 F_{12}$ will give a factor of two. Hence,

$$\delta S_{CS} = \frac{k}{2\pi} \int d^3x \delta A_0 F_{12} = \frac{k}{2\pi} \times \frac{2\pi}{\beta} \times \int_0^\beta d\tau \times \int_{S^2} d^2x F_{12} = 2\pi k.$$

So our action S is not gauge invariant! All is not lost because in the path integral what matters is e^{iS} . So, if we set k as an integer then our path integral is gauge invariant.

2.3 Degeneracy of Chern-Simons theory on a torus

The Chern-Simons action comes out as the infrared fix point of the effective field theory for lot of systems with topological order e.g. chiral spin states. We follow Wen [17] in this section.

$$S_{top} = \frac{k}{4\pi} \int_{\mathcal{M}} d^3x \epsilon^{\mu\nu\lambda} a_\mu \partial_\nu a_\lambda \quad (2.16)$$

Because the above equation has only linear terms in time derivatives, so the Hamiltonian is identically zero. Hence, all quantum states of the eq(2.16) have zero energy. Also the theory in eq(2.16) is a scaleless theory with no dimensional parameters. Fixing our gauge and setting $a_0 = 0$ and the corresponding E-L (Euler-Lagrange) constraint too, implies $F_{12} = 0$ and this fixes

$$a_1 = a_1(t), \quad a_2 = a_2(t). \quad (2.17)$$

Because of the residual gauge symmetries of the a'_i 's living on the torus of dimensions $L_1 \times L_2$. We have the gauge equivalent configurations given as:

$$(a_1, a_2) \sim (a'_1, a'_2) = \left(a_1 + \frac{2\pi n}{L_1}, a_2 + \frac{2\pi m}{L_2} \right), \quad (2.18)$$

where n and m are integers.

The gauge transformation that relates a_i to a'_i ($a'_i = a_i - iU^{-1}\partial_i U$) is given as

$$U = \exp \left[2\pi i \left(\frac{nx^1}{L_1} + \frac{mx^2}{L_2} \right) \right]. \quad (2.19)$$

Therefore, the gauge inequivalent configurations lie on a rectangle of size $2\pi/L_1 \times 2\pi/L_2$. We introduce the notation

$$(a_1, a_2) = \left(\frac{2\pi x}{L_1}, \frac{2\pi y}{L_2} \right). \quad (2.20)$$

Where now (x, y) and $(x+n, y+m)$ are gauge equivalent points parametrized on a torus. We can add a small mass terms to the action which generally comes from the rest of the part of effective field theory. Hence, the action now becomes

$$S = \int dt \left(k\pi(\dot{x}y - y\dot{x}) + \frac{m}{2}(\dot{x}^2 + \dot{y}^2) \right), \quad (2.21)$$

where later we can set the mass to be zero. The Lagrangian above describes the unit charged particle moving on a torus parametrized by (x, y) . The first term in the Lagrangian is of the form $\vec{v} \cdot \vec{A}$, which gives the uniform magnetic field $B = 2\pi k$ on the torus. The Hamiltonian of the eq(2.21) is given by

$$H = \frac{1}{2m} [-(\partial_x - iA_x)^2 - (\partial_y - iA_y)^2]. \quad (2.22)$$

This Hamiltonian has been studied by Haldane and Rezayi [18] and in the gauge: $A_x = 0, A_y = Bx = 2\pi kx$. It has been found that the ground states are given by

$$\psi_l(x, y) = \left[\sum_n \exp \left(2\pi(x + iy)(nk + l) - \frac{(nk + l)^2}{k} \pi \right) \right] e^{-Bx^2/2}, \quad (2.23)$$

where $l = 0, 1, \dots, k-1$. Hence, these **wavefunctions are k-fold degenerate**, where k was the level of the Chern-Simons theory. Similarly, for a *genus* $-g$ surface we have the degeneracy as $|k|^g$.

2.4 Wilson loops

The observables in a gauge theory should be gauge-invariant. The gauge-invariant objects in Chern-Simons theory are Wilson loops defined as follows:

$$W_i(C) = \exp\left(i \oint_{\gamma_i} dx_i^\mu A_\mu\right) \quad (2.24)$$

Another approach to see why the ground state should be k -fold degenerate is via Wilson loops. Along a torus we can consider two non-contractible loops (call it a and b). We can fix the gauge like above and also set $a_0 = 0$. Hence,

$$W(a) = \exp\left(i \oint_a a_1 \cdot dx\right) \quad \& \quad W(b) = \exp\left(i \oint_b a_2 \cdot dy\right). \quad (2.25)$$

On following canonical quantization one finds the conjugate momenta of a_1 is a_2 . Hence we also have

$$[a_1(\vec{x}), a_2(\vec{x}')] = \frac{2\pi i}{k} \delta(\vec{x} - \vec{x}'). \quad (2.26)$$

Using the Baker-Campbell-Hausdorff formula, one can easily see that

$$W(a)W(b) = W(b)W(a)e^{2\pi i/k}. \quad (2.27)$$

The lowest representation of this algebra is $|k|$ -dimensional and can correspond to below representation [6]:

$$\begin{aligned} W(b)|n\rangle &= |(n+1) \bmod |k|\rangle \\ W(a)|n\rangle &= e^{2\pi i n/k} |n\rangle \end{aligned} \quad (2.28)$$

where $n = 0, 1, \dots, |k| - 1$.

2.5 Linking number

If we have the source term of the form, $J = J_1 + J_2$ with

$$J_a^\mu = \oint_{\gamma_a} dx_a^\mu \delta^{(3)}(x - x_a(t)), \quad a \in \{1, 2\}. \quad (2.29)$$

Putting the above source term in eq(2.13), we get

$$A_\alpha^{\text{cl}}(x) = \frac{1}{2k} \sum_{a=1}^2 \oint_{\gamma_a} dx_a^\beta \epsilon_{\alpha\beta\mu} \frac{(x - x_a)^\mu}{|x - x_a|^3} \quad (2.30)$$

We want to calculate

$$\mathcal{Z}[J] = \langle W_1 W_2 \rangle = \exp(i\tilde{S}[A^{\text{cl}}]). \quad (2.31)$$

The above is true because the path integral action has quadratic terms. Substituting the value of A^{cl} in the above equation and removing self-interaction terms (or absorbing them in normalization constant), we get

$$\mathcal{Z}[J] = \exp\left(\frac{i}{2k} \oint_{\gamma_1} dx_1^\alpha \oint_{\gamma_2} dx_2^\beta \epsilon_{\alpha\beta\mu} \frac{(x - x_a)^\mu}{|x - x_a|^3}\right). \quad (2.32)$$

Hence,

$$\langle W_1 W_2 \rangle = \exp\left(\frac{2\pi i}{k} \Phi[\gamma_1, \gamma_2]\right); \quad (2.33)$$

where $\Phi[\gamma_1, \gamma_2]$ is an integer known as the *Gauss's linking integral* and is defined as follows:

$$\Phi[\gamma_1, \gamma_2] \equiv \frac{1}{4\pi} \oint_{\gamma_1} dx_1^\alpha \oint_{\gamma_2} dx_2^\beta \epsilon_{\alpha\beta\mu} \frac{(x - x_a)^\mu}{|x - x_a|^3}. \quad (2.34)$$

The above expression counts the number of times one curve winds around the other. An intuitive proof of the Gauss's linking number is sketched using the Ampere's law in the appendix A.3.

2.6 Non-abelian Chern-Simons theory

For non-abelian Chern-Simons theory we promote the gauge field A_μ to a matrix just like the Yang-Mills theory. The action for non-abelian Chern-Simons theory is given by [19]

$$S_{CS}[A] = \frac{k}{4\pi} \int_{\mathcal{M}} d^3x \epsilon^{\mu\nu\lambda} \text{Tr} \left[A_\mu \partial_\nu A_\lambda + \frac{2}{3} A_\mu A_\nu A_\lambda \right] \quad (2.35)$$

The proper gauge transformation for non-abelian case is

$$A_\mu \rightarrow U^{-1} A_\mu U + U^{-1} \partial_\mu U \quad (2.36)$$

and for the Wilson loops we need to introduce the path ordering in the exponential as shown below (which again makes them gauge invariant)

$$W_i(C) = \text{Tr} \left[\text{P} \exp \left(i \oint_{\gamma_i} dx_i^\mu A_\mu \right) \right]. \quad (2.37)$$

Just like for the abelian case we need the action to be gauge invariant, which here now transforms as

$$S_{CS} \rightarrow S_{CS} + 2\pi\nu k, \quad (2.38)$$

where

$$\nu = \frac{1}{24\pi^2} \int_{\mathcal{M}} d^3x \epsilon^{\alpha\beta\gamma} \text{Tr}[(U^{-1}\partial_\alpha U)(U^{-1}\partial_\beta U)(U^{-1}\partial_\gamma U)] \quad (2.39)$$

always turns out to be an integer known as the Pontryagin index. Hence, for this theory to be quantised the only constraint is k needs to be an integer. Since, this theory is a TQFT it's expectation value of Wilson loops has to be some invariants (like Eq (2.33)) which doesn't change as long as we don't change the topology of the Wilson loops. It was shown by Witten [19] that for a $SU(N)_k$, non-abelian theory with level- k on a S^3 manifold we get the Jones polynomial³ (Knot Invariant) for the below normalised expectation value of the Wilson loops

$$\text{Jones Polynomial of links} = \frac{\int_{S^3} \mathcal{D}a_\mu(x) W_{L_1} W_{L_2} \dots W_{L_n} e^{iS_{CS}}}{\int_{S^3} \mathcal{D}a_\mu(x) e^{iS_{CS}}}. \quad (2.40)$$

³See appendix A.2

3. Extrinsic defects in anyon models

In this section I would like to discuss a class of extrinsic defect called twist defect. Twist defects are interesting because even though they are not intrinsic topological quasiparticles, they show braiding very similar to non-abelian anyons (projective representation of the braid group). These defects provide us an alternative way to do topological quantum computing.

Twist defects [20] are extrinsic defects which are labelled by the symmetry g such that when a quasiparticle γ_i braids around it. It transforms into the quasiparticle $\gamma_{g(i)}$. The only condition on g is that under the transformation $\gamma_i \rightarrow \gamma_{g(i)}$, the fusion and braiding rules are preserved. The idea for twist defects here has been developed by examples in the sections below.

3.1 Twist defects in Toric Code

This section follows the work of H. Bombin [21]. We study twist defects on Toric code model by Wen [22], instead of the original spin-1/2 on links introduced by Kitaev [9]. The Hamiltonian is defined with spin-1/2 forming a square lattice, as:

$$H = - \sum_k A_k, \quad A_k = \sigma_k^x \sigma_{k+\mathbf{i}}^z \sigma_{\mathbf{i}+\mathbf{j}}^x \sigma_{k+\mathbf{j}}^z \quad (3.1)$$

where $k = (i, j)$, the spin index and $\mathbf{i} \equiv (1, 0)$; $\mathbf{j} \equiv (0, 1)$. The ground state is when all the A_k 's are equal to 1. Excitation here will be localized and gapped which can be obtained by exciting any of the plaquette $A_k = -1$. in which case we say that the plaquette k holds the quasiparticle.

To see clearly the quasiparticles we color the lattice as a checkerboard. To make the e and m excitations we take a site k and do $\sigma_k^z |\text{G.S.}\rangle$; doing this will flip A_k and $A_{k-\mathbf{i}-\mathbf{j}}$. This produces two local excitations which can be separated via string operators defined as $S_{\nwarrow, \swarrow} = \sigma_k^z$ and $S_{\nearrow, \searrow} = \sigma_k^x$. Note that one can only move the excited states freely along the diagonals. So, the excitations on dark plaquettes can move only along dark square and same thing for the light plaquettes. We label the quasiparticles living on dark (light)

plaquettes as the charge e (charge m)[†]. It's also easy to see that the string operators anticommute if they cross an odd number of times and they can be deformed as long as they don't go over an excitation. Using string operators we can also form loops: the dark loop γ^D , and the light loop γ^L .

One can observe that the closed string/loop operators commute with the plaquette operators and these can be used to describe the four types of quasiparticles as follows:

- If $S_{\gamma^D} = +1$ and $S_{\gamma^L} = +1$, then we have a total charge enclosed $\mathbf{1}$.
- If $S_{\gamma^D} = +1$ and $S_{\gamma^L} = -1$, then we have a total charge enclosed e .
- If $S_{\gamma^D} = -1$ and $S_{\gamma^L} = +1$, then we have a total charge enclosed m .
- If $S_{\gamma^D} = -1$ and $S_{\gamma^L} = -1$, then we have a total charge enclosed ϵ .

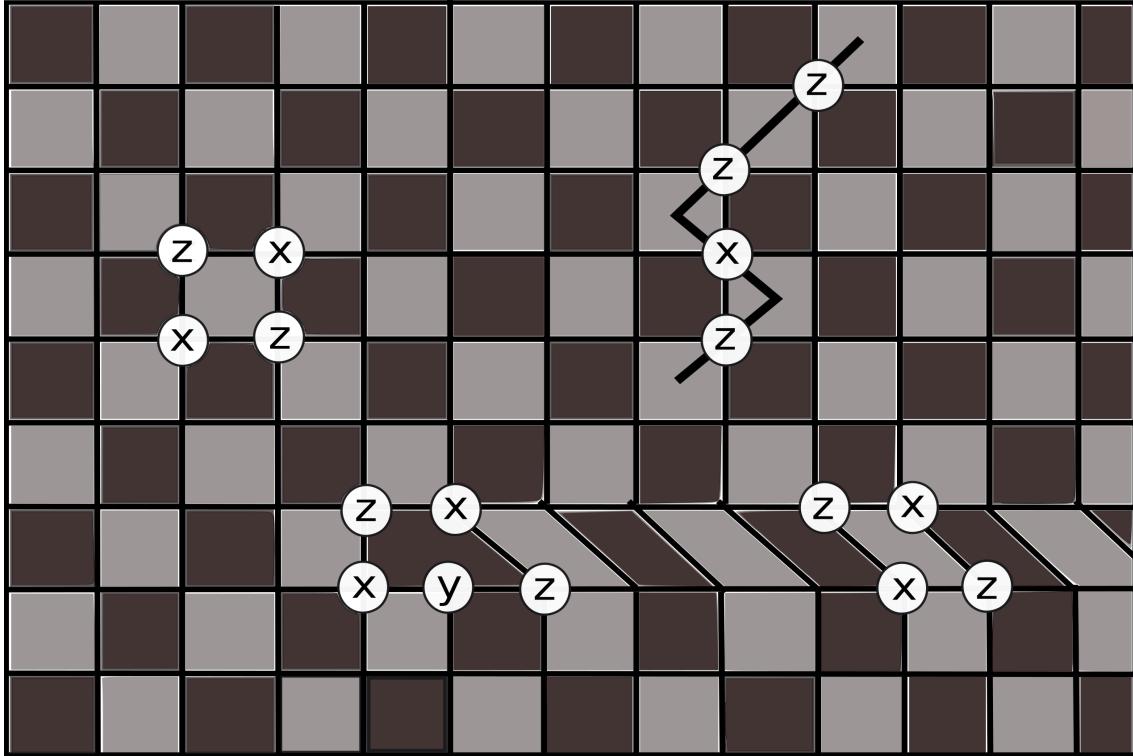


Figure 3.7: Shown in the figure are the normal plaquette terms, the plaquette term for the twist defect and the string operator creating two excitations on the light squares.

We introduce a twist defect as depicted in the picture fig(3.7). All the plaquette terms in the Hamiltonian still commute and so if we see far away from these twist defects we find that we have our earlier four quasiparticles. Near the twist defects we find that our e and m particles interchange if we go around our twist defect. Hence, ϵ can vanish near the twist and conversely pop out and so, we cannot distinguish between ϵ and $\mathbf{1}$ near

[†]Note: This choice was arbitrary. We can choose the opposite also.

twist. To distinguish excitations near the defect we cannot use the two types of strings as earlier. Instead we have a loop as shown in fig(3.8.a) which goes around both colors, but we have two different types of strings here which correspond to eigenvalues of string operator S_γ as $\pm i$. So, we distinguish two new topological charges as σ_+ for $S_\gamma = +i$ and σ_- for $S_\gamma = -i$. These charges now follow the below fusion statistics:

$$\begin{aligned}\sigma_\pm \times \sigma_\pm &= 1 + \epsilon, & \sigma_\pm \times \sigma_\mp &= e + m, \\ \sigma_\pm \times \epsilon &= \sigma_\pm, & \sigma_\pm \times e &= \sigma_\pm \times m = \sigma_\mp.\end{aligned}$$

A pictorial proof has been sketched in fig(3.8.b) for one of the fusion rules.

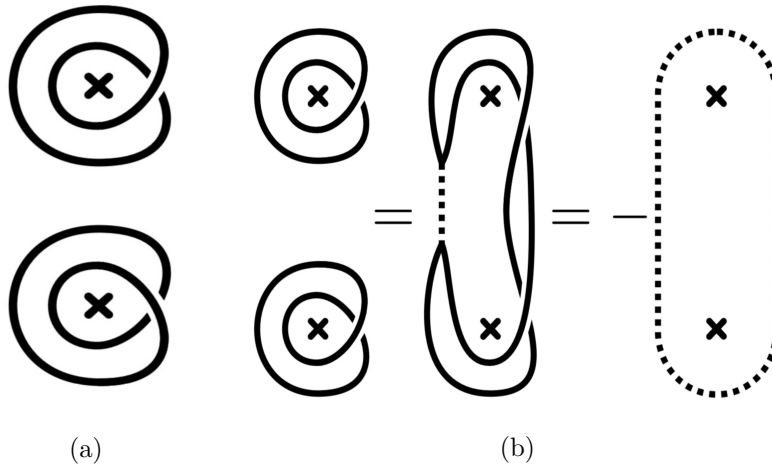


Figure 3.8: (a) Two different string operators or topological charges shown: σ_+ and σ_- (b) Fusion rule $\sigma_+ \times \sigma_+ = 1 + \epsilon$.

3.2 Genons in bilayer quantum Hall system

This section follows the work of Barkeshli [20]. It has been shown that topological nematic states (TNS) with Chern number N^4 , can be mapped to an N -layer quantum Hall system on a lattice and lattice dislocations on TNS can act as wormholes connecting different layers [23]. The Lagrangian⁵ can be written in terms of two gauge fields, a and $\tilde{a}$ as

$$\mathcal{L} = \frac{m}{4\pi}(a\partial a + \tilde{a}\partial\tilde{a}) \quad (3.2)$$

where m is an integer and $a\partial a \equiv \epsilon^{\mu\nu\rho}a_\mu\partial_\nu a_\rho$.

The action of twist defect in the system is like as soon as the particle crosses the defect in one layer, it transports to the other layer. Hence, if we take our manifold for fields to

⁴For our discussion here we take $N = 2$ and work with the bilayer system.

⁵One can also add the term $\frac{l}{4\pi}(a\partial\tilde{a} + \tilde{a}\partial a)$ in the Lagrangian which causes everywhere $|m|$ to be replaced by $|m - l|$.

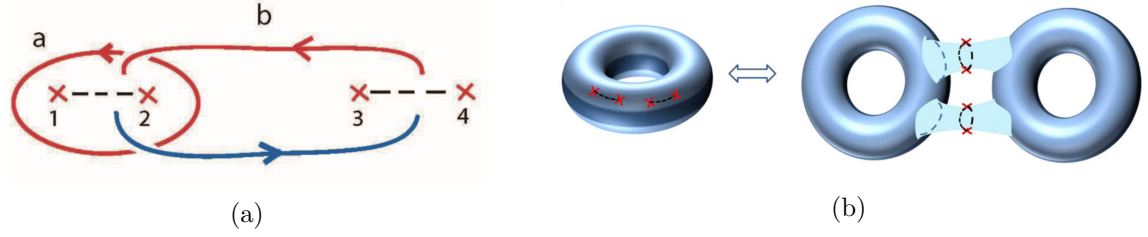


Figure 3.9: (a) A pair of twist defect for $(mm0)$ states on a sphere has only two non-contractible loops labelled a and b. (b) Change in genus from 2 holes of two tori to 3 holes because of a pair of twist defects. Images taken from [20, 23].

be the surface of the sphere, then two pairs of twist defect produce two non-contractible loops as shown in fig(3.9.a). These loops satisfy eq (2.27) as they cut only once. Hence, the presence of twist defects made our Hilbert space m -dimensional which without its presence was one-dimensional. Similarly one can see that we have a change in genus because of the twist defects (as shown in fig(3.9.b)). One can generalise these for n pairs of twist defects, we will have $2(n - 1)$ non-contractible loops which act like $n - 1$ copies of the algebra eq(2.27). Hence, the Hilbert space will be $|m|^{n-1}$ dimensional, which gives the quantum dimension of each twist defect as $\sqrt{m}$.

So, we can map our theory on sphere with n -pairs of twist defect to a *genus* $-(n - 1)$ theory. These twist defects are then called *genons*. The action of braiding these twist defects can be seen from fig(3.10) that

$$B_{12} : b \rightarrow b + a \quad (3.3)$$

and

$$B_{23}^\dagger : a \rightarrow b + a. \quad (3.4)$$

Hence, braiding of twist defects corresponds to Dehn twists in the effective *genus* g surface. In the diagonal basis of eq(2.28) one can calculate the braiding matrix and finds that B_{12} is diagonal in that basis and is given as [20, 19, 24]

$$B_{12}|n\rangle = e^{i\theta} e^{i\pi n^2/m + i\pi n m} |n\rangle \quad (3.5)$$

and the B_{23} matrix is given as

$$(B_{23})_{\alpha\delta} = \frac{e^{i\phi}}{|m|} \sum_{\beta=0}^{|m|-1} e^{i\pi\beta m + i\pi(\beta^2 + 2\beta(\alpha - \delta))/m} \quad (3.6)$$

where the overall phases θ and ϕ depends on non-universal details of the path.

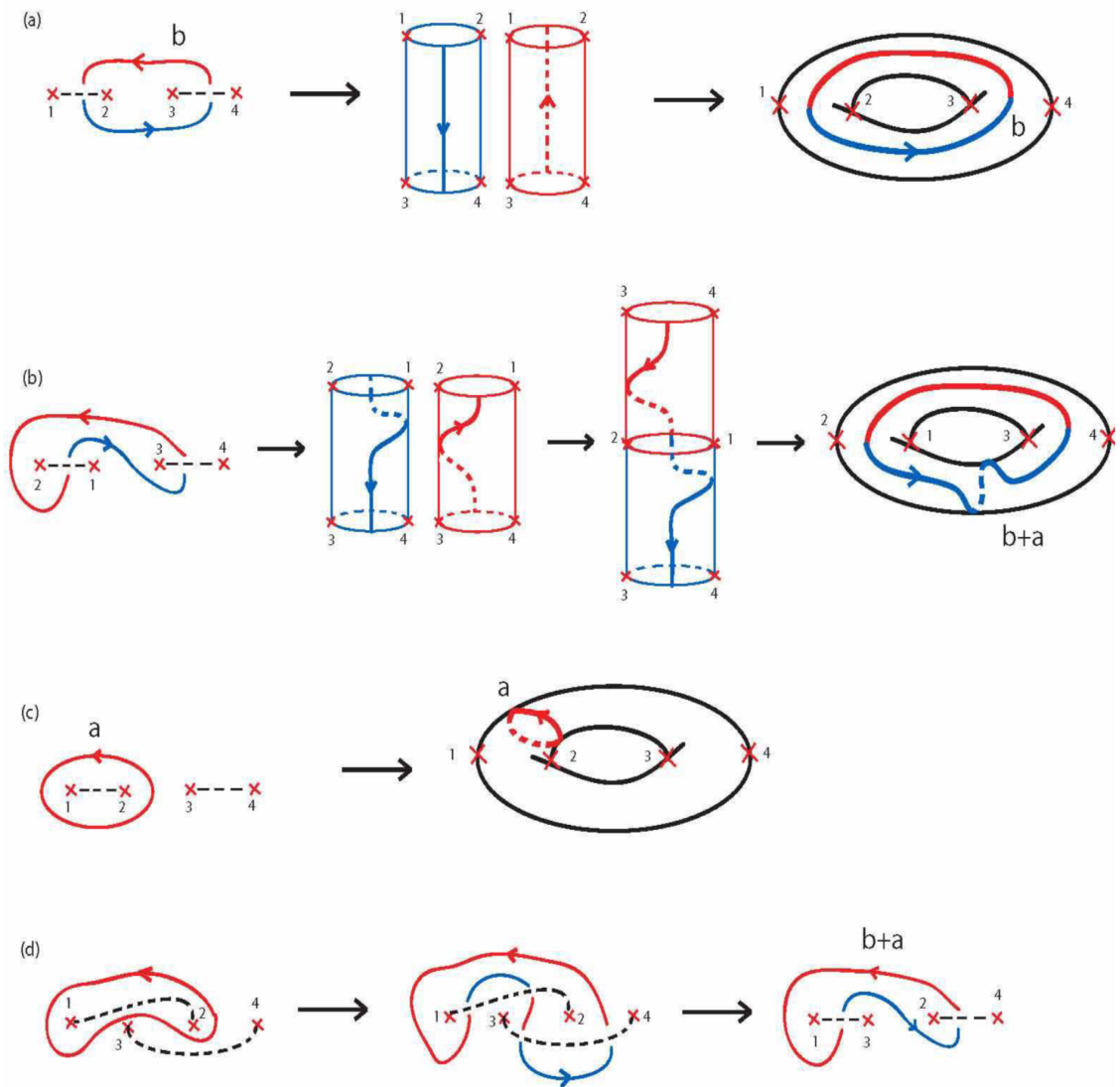


Figure 3.10: Braiding the twist defects produces Dehn twist in *genus* g surface as is evident. Image taken from [20].

Appendix A:

Categories, Knots & Links

A.1 Categories, Functors & Natural transformations

A **category**[†] $\mathcal{C}$ consists of the following data:

- *Objects*, denoted by $A, B, \dots$
- *Morphisms* between objects denoted by arrows and labelled by $f, g, \dots$
- *Composition of morphisms* : If $f : A \rightarrow B$ and $g : B \rightarrow C$, then $g \circ f : A \rightarrow C$
- *Identity morphism* for all the objects, i.e, $1_A : A \rightarrow A$

The morphisms have to follow the following rules:

- **Associative rule** in the morphisms, i.e., If $f : A \rightarrow B$, $g : B \rightarrow C$ and $h : C \rightarrow D$, then $f \circ (g \circ h) = (f \circ g) \circ h$.
- **Identity rule** for the identity morphisms, i.e, If $f : A \rightarrow B$ then $f \circ 1_A = f = 1_B \circ f$

Note: There can exist multiple morphisms between two objects of the category. It's not necessary to have arrow both ways between any two objects (example: a category of 2 objects with only one non-identity arrow). Our goal in category theory is to study about morphism and their properties, so, we do not care much about our objects. An example of category could be objects whose morphisms are labelled by elements of some group, and composition of morphisms is equal to group element products. It's easy to check that this category is associative and follows the identity rule, where identity morphism is labelled by identity of the group. Standard examples of categories are **Set** (objects are sets and arrows are functions) and **Vect** (objects are Vector spaces and arrows are linear

[†]The reference for this Appendix is Richard Borchers introductory videos in Category theory on YouTube and Bartlett Master's thesis [25].

operator over some field k).

A map between two categories, $\mathcal{A} \xrightarrow{F} \mathcal{B}$ is called a **functor** if the map preserves:

- If $f : a \rightarrow b$ where a and b are objects in $\mathcal{A}$, and $F(f) : F(a) \rightarrow F(b)$ where $F(a)$ and $F(b)$ are objects in $\mathcal{B}$. Then for any two composable morphisms f and g in $\mathcal{A}$

$$F(f \circ g) = F(f) \circ F(g)$$

- For all objects a in $\mathcal{A}$, $F(\mathbf{1}_a) = \mathbf{1}_{F(a)}$

In category theory, isomorphism of categories is not a much used term because it is quite a forcing condition and these things don't come up in our generally used category theory. Instead, we use the idea of natural transformations and natural isomorphisms here.

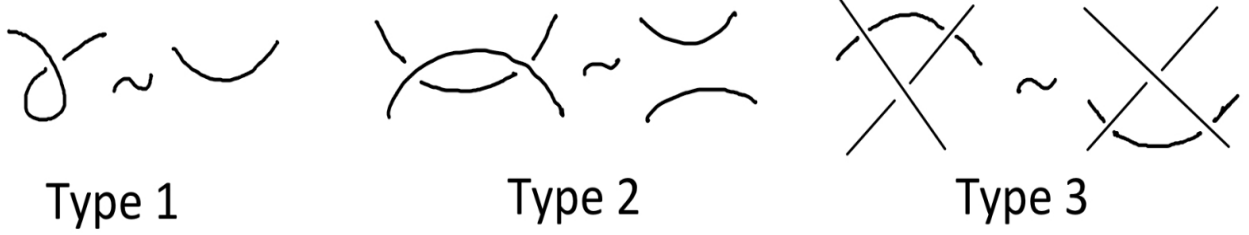
Let there be two categories $\mathcal{A}$ and $\mathcal{B}$ with two functors F and G defined on them. Then the **natural transformation** is a collection of morphisms $\{\alpha_a : F(a) \rightarrow G(a)\}$ indexed by objects of $\mathcal{A}$, such that for any morphism $a \xrightarrow{f} b$ in $\mathcal{A}$, the below square commutes in $\mathcal{B}$:

$$\begin{array}{ccc} F(a) & \xrightarrow{\alpha_a} & G(a) \\ F(f) \downarrow & & \downarrow G(f) \\ F(b) & \xrightarrow{\alpha_b} & G(b) \end{array}$$

$$\text{i.e., } G(f)(\alpha_a(F(a))) = \alpha_b(F(f)(F(a))) = G(b) \quad (\text{A.7})$$

A.2 Knots and Knot invariant polynomials

The study of Knot theory has been primarily the work of Alexander, Jones, Kauffman and Conway. They have studied knots and defined knot invariants (a quantity if not equal for two knots then they are not isotopic). For knots to have isotopy we define three kinds of allowed moves which are known as the Reidmeister moves. They are shown in the figure below:



We have a polynomial associated with links called the Kauffman bracket polynomial defined as follows:

- $\langle \bigcirc \rangle = 1$;
- $\langle L \cup \bigcirc \rangle = (-A^2 - A^{-2}) \langle L \rangle$
- $\langle \text{crossing} \rangle = A \langle \text{smooth} \rangle + A^{-1} \langle \text{smooth} \rangle$

One can check two isotopic links have the same Kauffman bracket unless we do a Type 1. Reidmeister move. To correct this problem we assign direction/arrows to the links and define the below two signs for two types of crossings.



We then calculate the writhe $w(L)$, which is just the total sum of the all the crossings.

One can verify that now the quantity

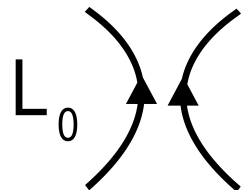
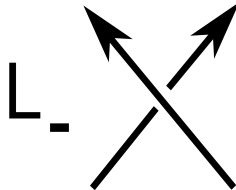
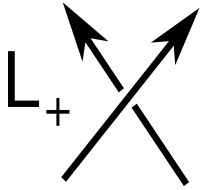
$$X(L) = (-A)^{-3w(L)} \langle L \rangle, \quad (\text{A.8})$$

is invariant under all types of Reidmeister moves. Now, the Jones polynomial is defined as,

$$V(L) = X(L)_{A=q-1/4} \quad (\text{A.9})$$

The Jones polynomial of a trivial knot is one. It's easy to verify that the Jones polynomial obey *Skein relation*:

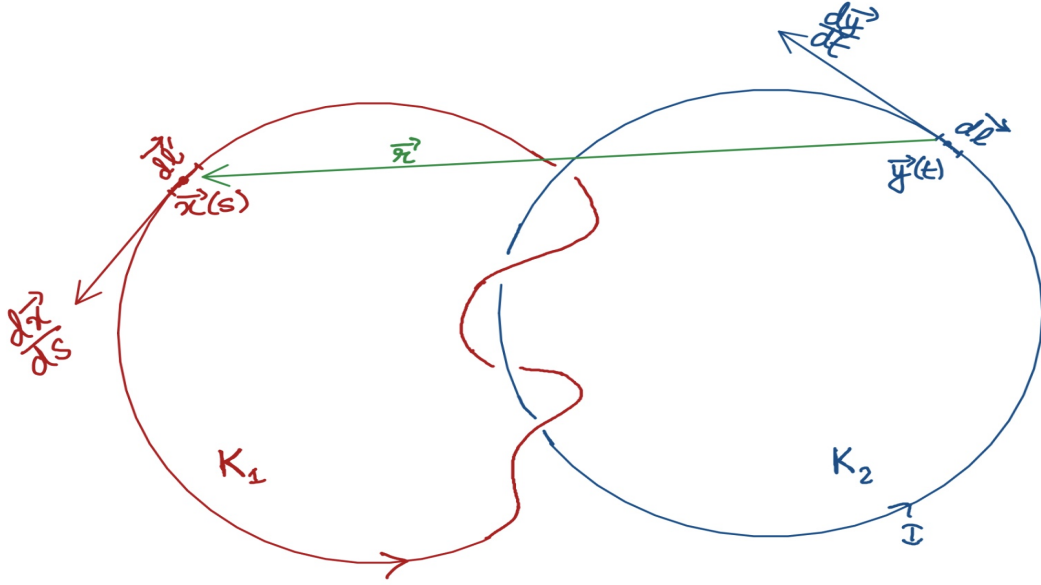
$$q^{-1}V(L_+) - qV(L_-) + (q^{-1/2} - q^{1/2})V(L_0) = 0 \quad (\text{A.10})$$



A.3 Gauss's linking number

For two link K_1 and K_2 as shown in the figure, the link invariant is given as:

$$\begin{aligned} \text{Link}(K_1, K_2) &= \int_{K_1 \times K_2} \frac{d\vec{x}}{ds} \times \frac{d\vec{y}}{dt} \cdot \frac{\vec{x} - \vec{y}}{4\pi|\vec{x} - \vec{y}|^3} ds dt \\ &= \oint_{K_1} dx^\mu \oint_{K_2} dy^\nu \epsilon^{\mu\nu\rho} \frac{(x - y)^\rho}{4\pi|x - y|^3} \end{aligned} \quad (\text{A.11})$$



As shown in the figure take two loops which are linked with one another. We calculate the line integral of magnetic field due to second loop (carrying a current “I”) around the first loop. This can be written using Ampere’s law as

$$\oint_{K_1} \vec{B} \cdot d\vec{l} = \mu_0 \times [\text{Linking number}] \times I = \mu_0 I \oint_{K_1} \oint_{K_2} \frac{1}{r^3} [(d\vec{l} \times d\vec{l}') \cdot \vec{r}]. \quad (\text{A.12})$$

$$\Rightarrow \text{Linking number} = \oint_{K_1} \oint_{K_2} \frac{1}{r^3} [(d\vec{l} \times d\vec{l}') \cdot \vec{r}] = \oint_{K_1} dx^\mu \oint_{K_2} dy^\nu \epsilon^{\mu\nu\rho} \frac{(x - y)^\rho}{4\pi|x - y|^3} \quad (\text{A.13})$$

Appendix B:

Calculations

B.1 Chern-Simons classical solution for Lorentz gauge

We have

$$\frac{k}{2\pi}\epsilon^{\mu\nu\rho}\partial_\nu A_\rho = J^\mu \quad \text{and} \quad \partial_\mu A^\mu = 0 \quad (\text{B.14})$$

and we need to solve for A^μ 's. Taking derivative of J^μ and summing those by $\epsilon_{\alpha\beta\mu}$ which then gives the α component of the sum as

$$\epsilon_{\alpha\beta\mu} \partial^\beta J^\mu = \frac{k}{2\pi}\epsilon^{\alpha\beta\mu}\epsilon^{\mu\nu\rho}\partial_\beta\partial_\nu A_\rho. \quad (\text{B.15})$$

Using the identity

$$\epsilon^{ijk}\epsilon^{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}. \quad (\text{B.16})$$

$$\implies \epsilon_{\alpha\beta\mu} \partial^\beta J^\mu = \frac{k}{2\pi}(\delta_{\beta\rho}\delta_{\alpha\nu} - \delta_{\beta\nu}\delta_{\alpha\rho})\partial_\beta\partial_\nu A^\rho \quad (\text{B.17})$$

Using Lorentz gauge condition, one of the delta term vanishes and the equation becomes

$$(\partial_t^2 + \partial_x^2 + \partial_y^2)A_\alpha = -\frac{2\pi}{k}\epsilon_{\alpha\beta\mu} \partial^\beta J^\mu \quad (\text{B.18})$$

Now since the three-dimensional Green's function for Laplace equation is

$$G(\vec{x}, \vec{x}') = -\frac{1}{4\pi|\vec{x} - \vec{x}'|}. \quad (\text{B.19})$$

Hence,

$$A_\alpha = \frac{1}{2k} \int d^3y \frac{\epsilon_{\alpha\beta\mu} \partial^\beta J^\mu}{|x - y|} \quad (\text{B.20})$$

B.2 Quantum dimension and N matrices

For finding the quantum dimension of the particle a, we take 5 such particles and fix the final result of fusion say to be e. Then

$$\text{dimension of fusion} = \sum_{b,c,d} N_{aa}^b N_{ba}^c N_{ca}^d N_{da}^e \quad (\text{B.21})$$

$$\implies \text{dimension of fusion} = [N_a^4]_a^e \quad (\text{B.22})$$

Similarly, for n anyons, the dimension of Hilbert space is $[N_a^{n-1}]_a^e$. We can diagonalise N_a and write it as $U\Lambda U^{-1}$. This implies

$$[N_a^{n-1}]_a^e = U\Lambda^{n-1}U^{-1}. \quad (\text{B.23})$$

For finding the quantum dimension we set n to be very large; in that case all other eigenvalues give very small value when powered to n^{th} power in comparison to the largest eigenvalue. Hence, only diagonal entry which contributes in Λ is the largest eigenvalue and all others can be taken to zero. Thus all elements of N_a^{n-1} are proportional to the largest eigenvalue powered to n. Hence, the quantum dimension of any anyon type a is the largest eigenvalue of the N_a matrix.

Since, $(a \times b) \times c = a \times (b \times c)$. Thus,

$$\sum_d N_{ab}^d N_{cd}^e = \sum_f N_{cb}^f N_{af}^e. \quad (\text{B.24})$$

This implies that any two matrices N_a and N_c commute, i.e., $[N_a, N_c] = 0$. Hence all N matrices can be diagonalised by the same unitary matrix U. It turns out U is the modular S-matrix for this anyon theory.

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